PORTFOLIO PROJECT | DATA ANALYTICS

# Financial Loan Analysis

Insights Report – Growth, Risk and Profitability (2021)

## 1. Executive Summary

This report analyses **38,576 personal loans issued in 2021**. The lender put out **$435.8M** and has received **$473.1M** so far. The aim is to understand how the portfolio is growing, where the risk sits, and which loans make money.

| 38.6K | $435.8M | $473.1M | 12.05% | 13.33% |
|---|---|---|---|---|
| Loan Applications | Funded Amount | Received Amount | Avg Interest Rate | Avg DTI |

| 86.18% | 13.82% | −$28.2M | +$32.0M | 7.7% |
|---|---|---|---|---|
| Good Loans | Bad Loans | Lost on Charged-off Loans | Net Profit (finished loans) | Return on Money Lent |

### Five things the data tells us

1. **Growth is fast, and it is getting riskier.** Applications grew 85% and money lent grew 116% from January to December. The average loan rose from about $10,700 to $12,500 and the average interest rate from 11.5% to 12.4%.
2. **Bad loans are costly.** 5,333 loans were charged off and the lender recovered only $37.3M of $65.5M lent. This loss wipes out about **47% of the $60.2M profit** earned from fully paid loans.
3. **Risk is predictable.** Charge-offs rise from 5.7% (grade A) to 33.1% (grade G). They are also much higher for 60-month loans (25.0% vs 10.7%), small business loans (26.7%), loans above $20K (20.0%) and borrowers earning under $30K (17.6%).
4. **Profit comes from the middle.** Grades B and C produced about half of all profit ($16.4M of $32.0M). Grade A is safe but earns only 4.5%, and **small business is the only purpose that loses money (−2.8%)**.
5. **Some common checks do not reduce risk.** Verified borrowers default more (15.6%) than non-verified ones (12.4%), and employment length shows no clear pattern. The portfolio is also concentrated: four states hold 42% of the lending.

## 2. Data and Method

The data has 38,576 loans and 24 columns covering January to December 2021. In Python, the data was checked for duplicates (none) and missing values (only employee title, 1,438 rows, filled as “Unknown” so that all loans are kept). Dates and text columns were fixed, and **annual income outliers (1,824 loans, 4.7%) were capped** at $145,751 so that a few very high incomes would not distort the charts.

Loans that are **Current** (1,098) are still being repaid, so their results are not final. For risk and profit analysis only **finished loans (37,478)** are used. **Charge-off rate** = charged-off loans ÷ finished loans, which is 14.2% overall (the dashboard’s 13.82% divides by all loans including Current).

## 3. Dashboard

The Power BI dashboard has two pages, built for managers who want to monitor the portfolio at a glance. Both pages share the same five KPI cards, each with a **month-to-date (MTD)** value for December and **month-over-month (MoM)** change versus November. Filters for **Grade, Purpose and State** apply to all visuals.

### Page 1 – Overview

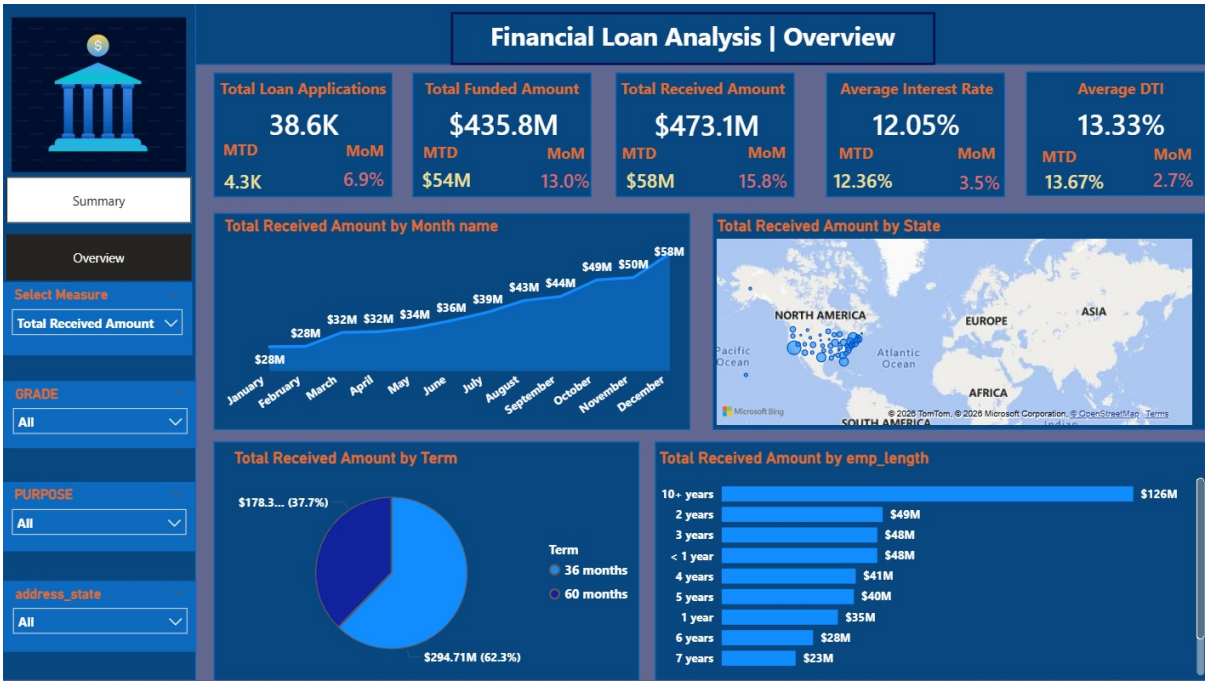
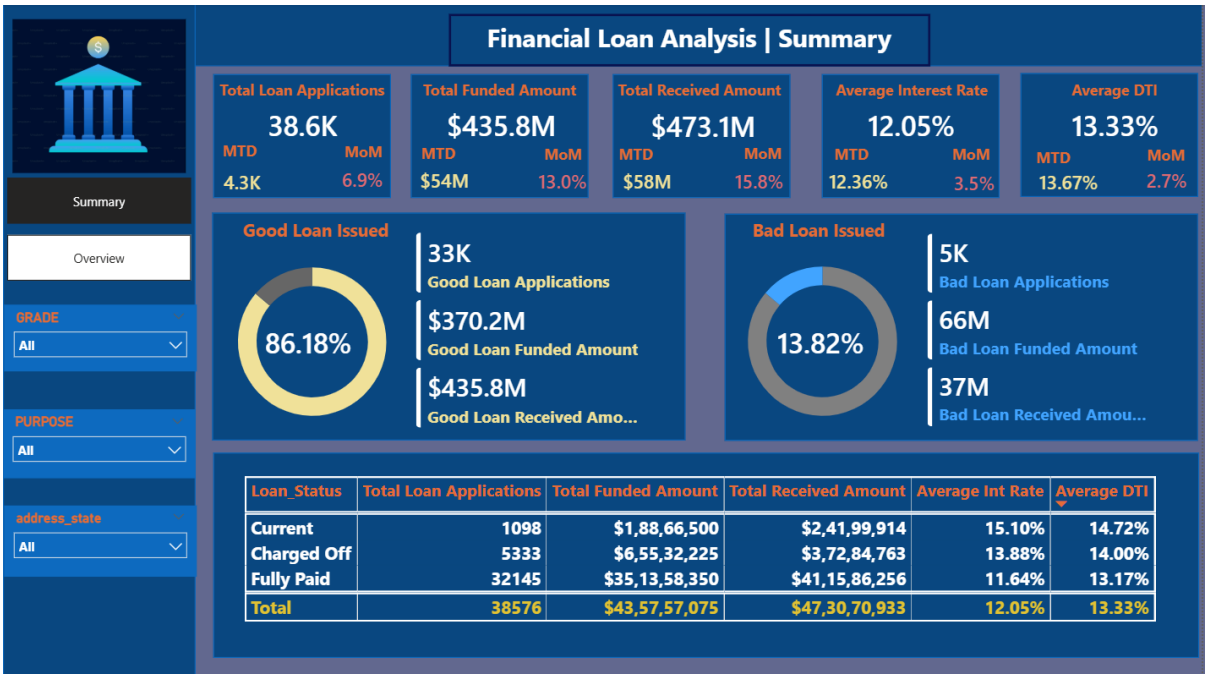



*Figure 1: Dashboard Page 1 – Overview*

- **Total Received Amount by Month:** rises steadily from $28M in January to $58M in December.
- **Total Received Amount by State (map):** the biggest bubbles are on the US West Coast and East Coast; California is the largest market.
- **By Term (pie):** 36-month loans bring in 62.3% ($294.7M) and 60-month loans 37.7% ($178.4M).
- **By Employment Length (bar):** borrowers with 10+ years of employment bring in the most ($126M), far ahead of every other group.
- **Select Measure** slicer: switches the charts between measures, so one page can answer several questions.

## Page 2 – Summary




| Loan_Status | Total Loan Applications | Total Funded Amount | Total Received Amount | Average Int Rate | Average DTI |
|---|---|---|---|---|---|
| Current | 1098 | $1,88,66,500 | $2,41,99,914 | 15.10% | 14.72% |
| Charged Off | 5333 | $6,55,32,225 | $3,72,84,763 | 13.88% | 14.00% |
| Fully Paid | 32145 | $35,13,58,350 | $41,15,86,256 | 11.64% | 13.17% |
| Total | 38576 | $43,57,57,075 | $47,30,70,933 | 12.05% | 13.33% |

*Figure 2: Dashboard Page 2 – Summary*

- **Good Loan Issued: 86.18%** – about 33K loans and $370.2M funded.
- **Bad Loan Issued: 13.82%** – about 5K loans, $66M funded and only $37M received back.
- **Loan Status table:** Charged Off loans have a higher average interest rate (13.88%) and DTI (14.00%) than Fully Paid loans (11.64% and 13.17%), the first clue that riskier borrowers are being charged more.

> **Dashboard check:** every KPI on both pages was rebuilt in Python and matches exactly (38,576 loans, $435.8M funded, $473.1M received, 12.05% rate, 13.33% DTI, and the December MoM changes of 6.9%, 13.0%, 15.8%, 3.5% and 2.7%).

# 4. Insights from the Python Analysis

## 4.1 Portfolio growth




*Figure 3: Monthly loan applications (2021)*

**What it shows:** Applications grew almost every month, from 2,332 in January to 4,314 in December (+85%). February is the only dip.

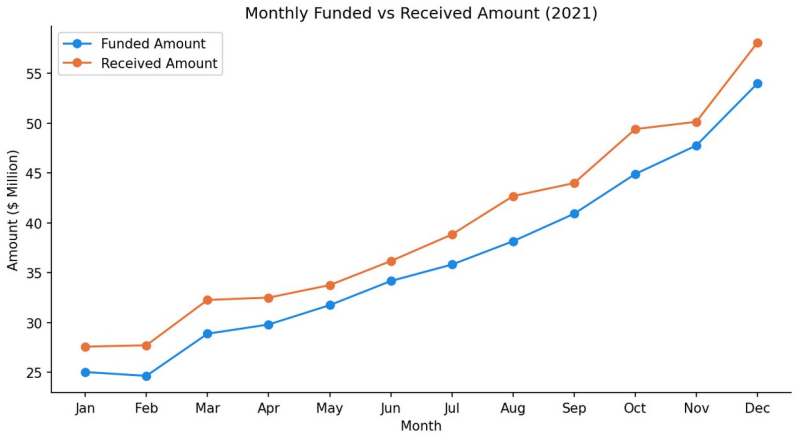



*Figure 4: Monthly funded vs received amount*

**What it shows:** Money lent grew from $25.0M to $54.0M (+116%), faster than the number of loans. Received amount stays above funded every month because repayments include interest.

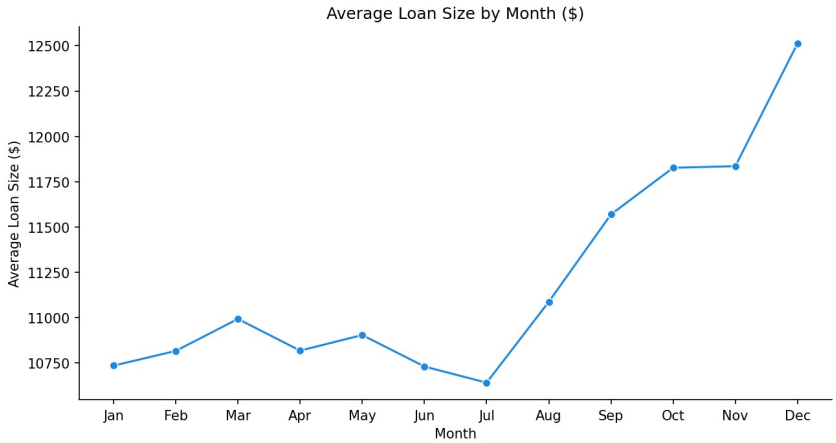



*Figure 5: Average loan size by month*

**What it shows:** The average loan grew from about $10,700 in January to $12,500 in December (+17%).

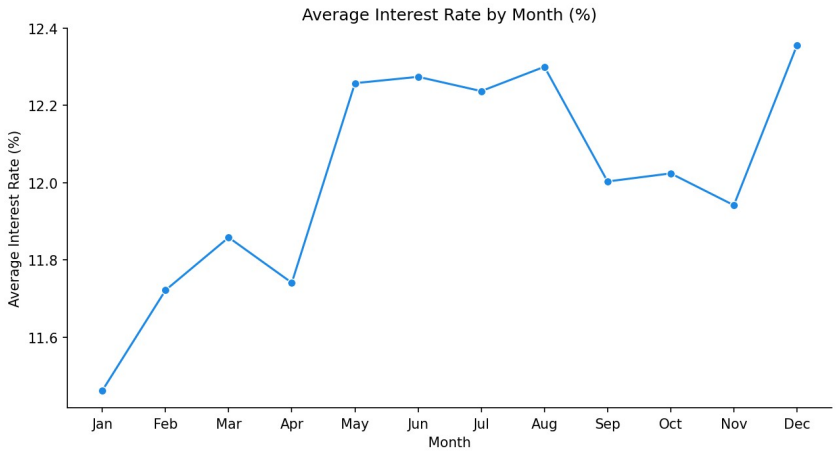



*Figure 6: Average interest rate by month*

**What it shows:** The average rate climbed from 11.5% to 12.4%. Growth is coming from bigger and more expensive loans – a rise that needs monitoring because larger loans are riskier (see 4.3).

## 4.2 Who the borrowers are

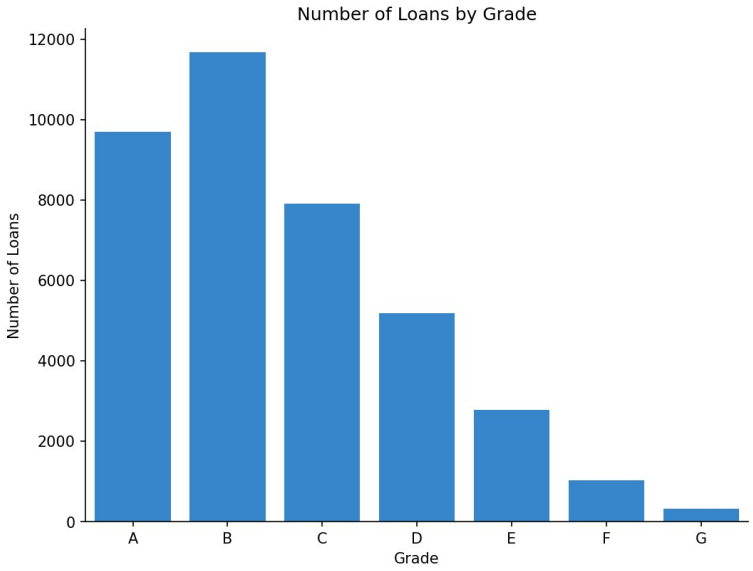



*Figure 7: Number of loans by grade*

**What it shows:** Grade B is the largest group (30.3% of loans), followed by A (25.1%). Grades A and B are 55% of all loans; grades D–G are about 24% of loans but 31% of the money lent.

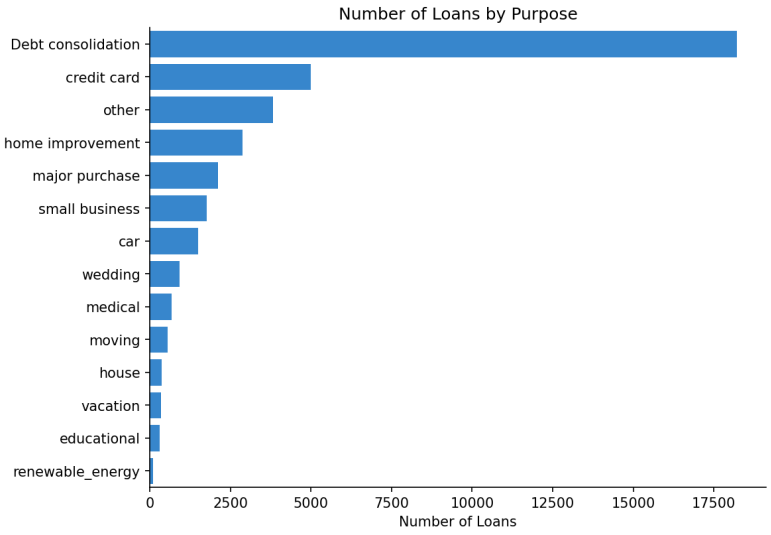



*Figure 8: Number of loans by purpose*

**What it shows:** Debt consolidation is 47% of loans and 53% of the money lent. With credit card refinancing, about 60% of loans are used to repay existing debt.

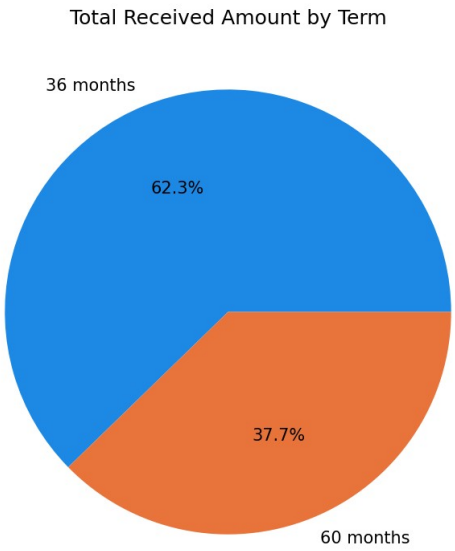



*Figure 9: Total received amount by term*

**What it shows:** 36-month loans are 73% of loans and bring in 62.3% of the money received; 60-month loans bring in 37.7%.

## 4.3 Loan quality and risk

Charge-off rate is the share of finished loans that were written off. Overall it is **14.2%**.

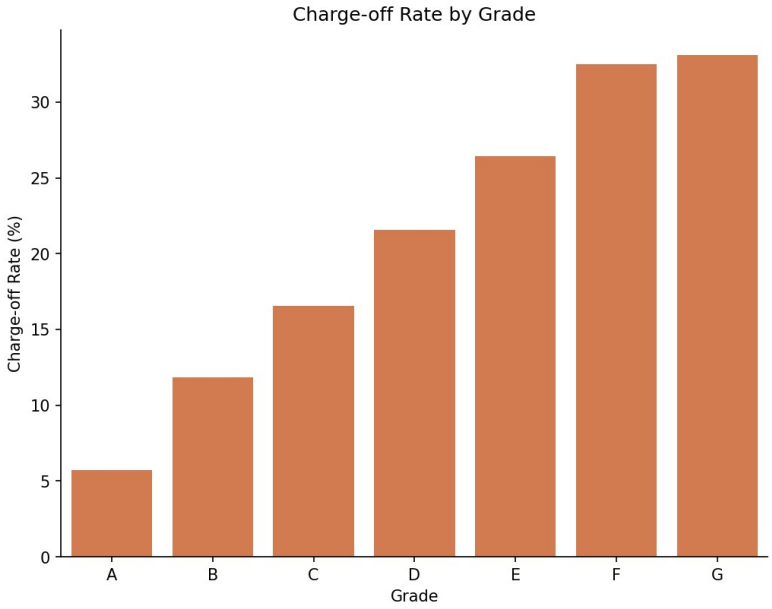



*Figure 10: Charge-off rate by grade*

**What it shows:** Risk increases with every grade, from 5.7% (A) to 33.1% (G). A grade F or G borrower is about six times more likely to default than a grade A borrower.

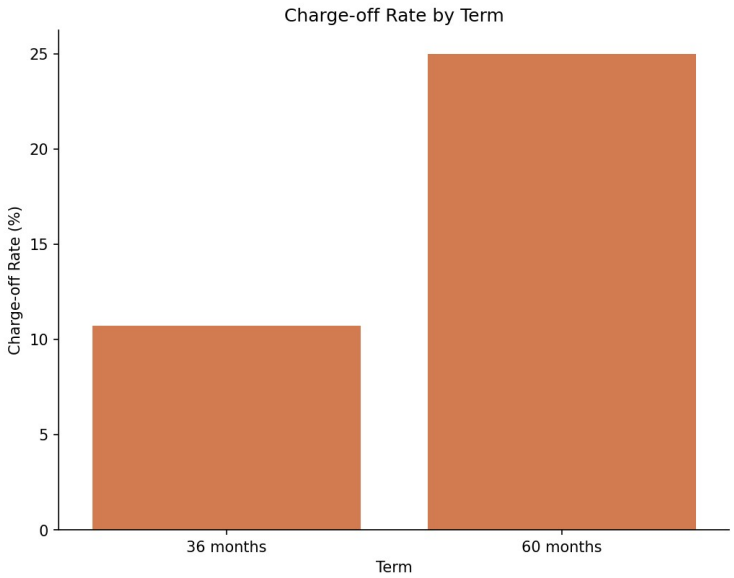



*Figure 11: Charge-off rate by term*

**What it shows:** 60-month loans are charged off 25.0% of the time against 10.7% for 36-month loans – more than double.

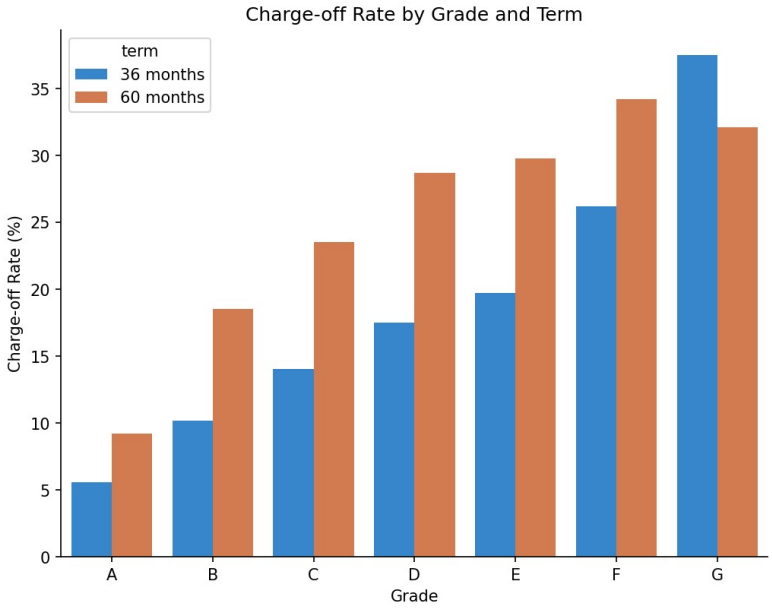



*Figure 12: Charge-off rate by grade and term*

**What it shows:** The 60-month penalty applies in every grade from A to F, so it is not just a grade effect. For example, grade C goes from 14.0% (36 months) to 23.5% (60 months).

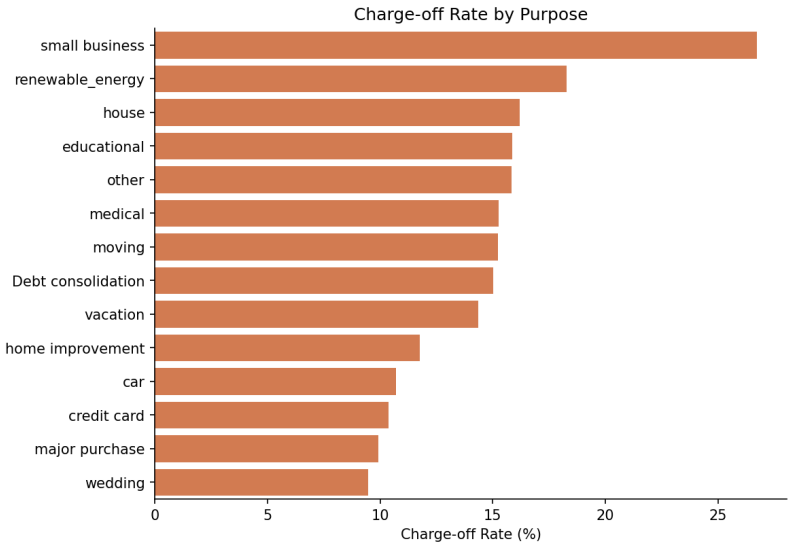



*Figure 13: Charge-off rate by purpose*

**What it shows:** Small business loans are by far the riskiest at 26.7%. The safest purposes are wedding (9.5%), major purchase (9.9%), credit card (10.4%) and car (10.7%).

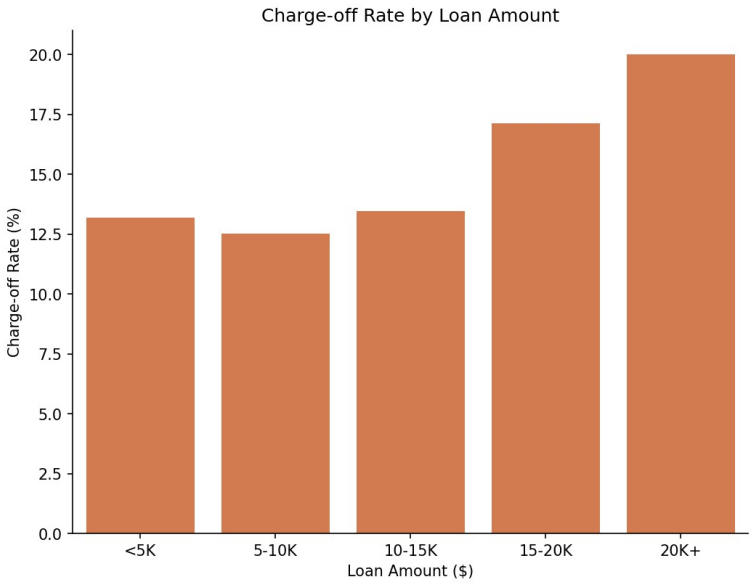



*Figure 14: Charge-off rate by loan amount*

**What it shows:** Loans under $15K have charge-off rates of 12.5%–13.5%. The rate jumps to 17.1% for $15–20K and 20.0% above $20K.

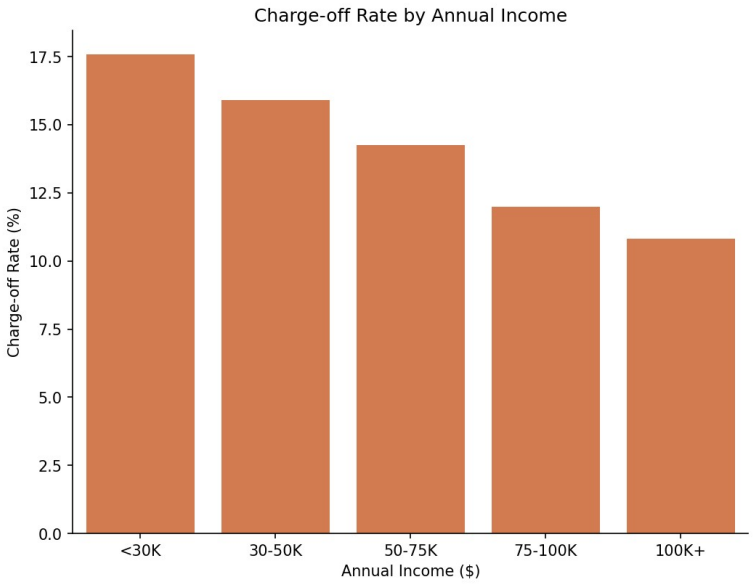



*Figure 15: Charge-off rate by annual income*

**What it shows:** Risk falls steadily as income rises: 17.6% for incomes under $30K down to 10.8% above $100K.

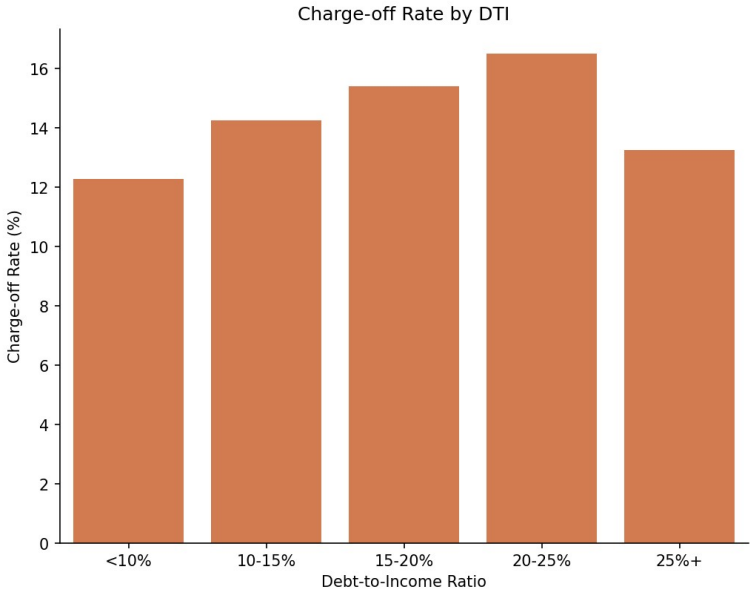



*Figure 16: Charge-off rate by DTI*

**What it shows:** Risk rises with DTI from 12.3% (below 10%) to 16.5% (20–25%). The 25%+ group has only 596 loans, so its lower rate is not reliable.

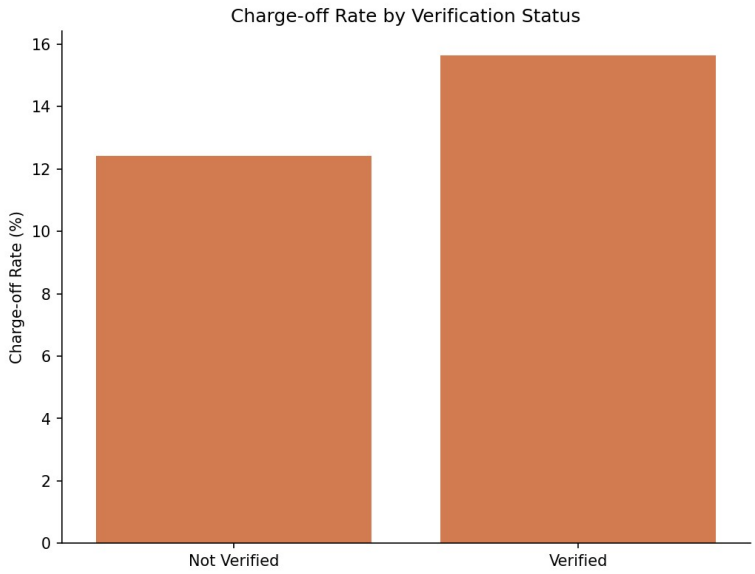



*Figure 17: Charge-off rate by verification status*

**What it shows:** Verified borrowers default more (15.6%) than non-verified (12.4%) – they also pay higher rates (12.5% vs 11.2%), so the lender may be verifying the riskier applicants.

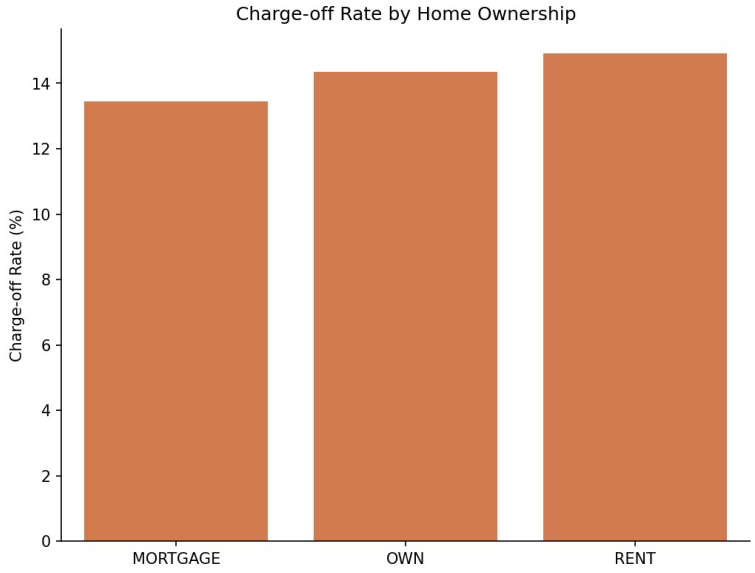



*Figure 18: Charge-off rate by home ownership*

**What it shows:** Home ownership has only a small effect: Mortgage 13.5%, Own 14.3%, Rent 14.9%.

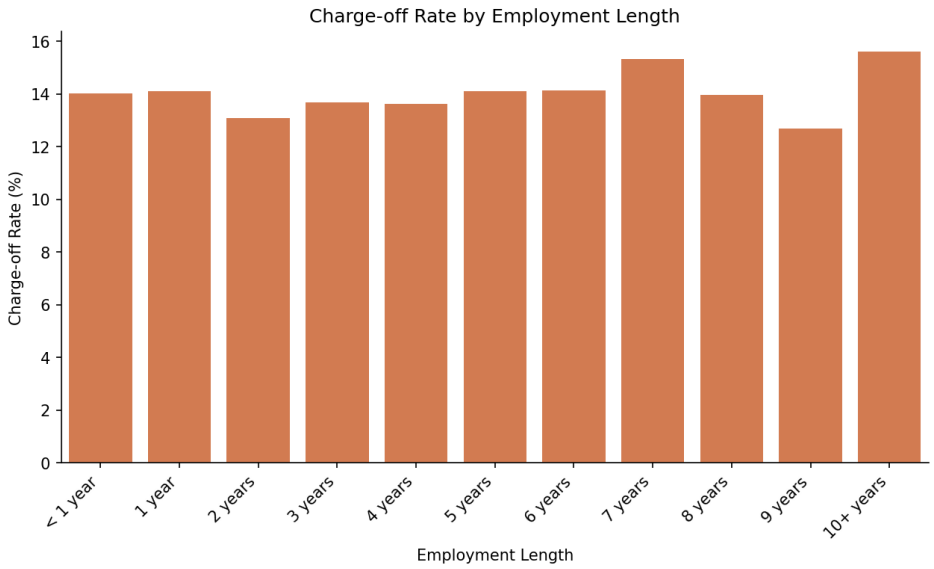



*Figure 19: Charge-off rate by employment length*

**What it shows:** Every group sits between 12.7% and 15.6% with no clear pattern, so employment length is a weak signal for approval decisions.

## 4.4 Profitability

Profit is the money received minus the money lent, for finished loans only. The return is a simple return and is not adjusted for how long the money was lent.

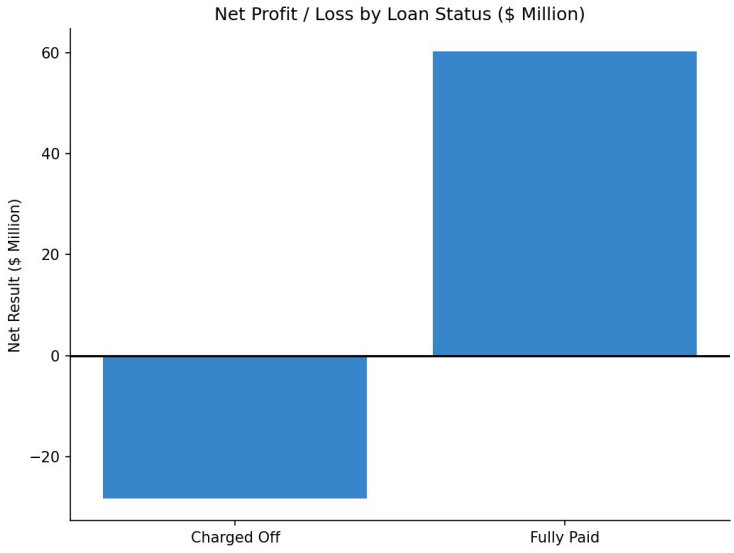



*Figure 20: Net profit / loss by loan status*

**What it shows:** Fully paid loans earned +$60.2M, while charged-off loans lost $28.2M (only 56.9% of the money lent was recovered). Net profit is $32.0M, a 7.7% return – charge-offs erase about 47% of the profit from good loans.

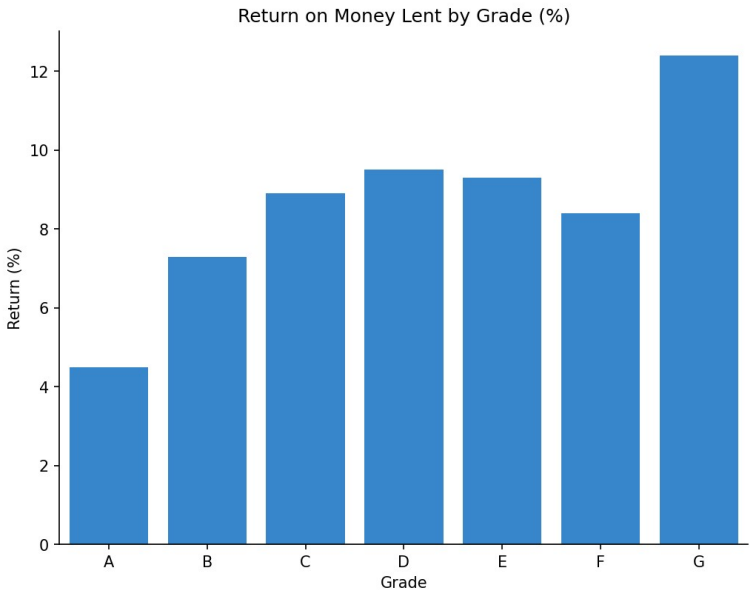



*Figure 21: Return on money lent by grade*

**What it shows:** Grade A is the safest but returns only 4.5%. Grades D and E return 9.5% and 9.3% – their higher interest more than pays for their higher defaults.

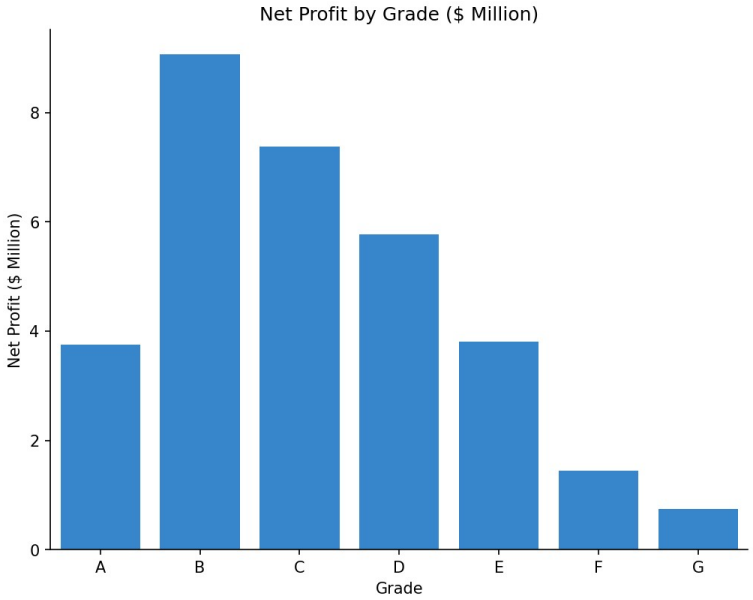



*Figure 22: Net profit by grade*

**What it shows:** Grades B and C earn the most profit ($9.1M and $7.4M); grade A makes only $3.8M. F and G are too small to matter ($1.5M and $0.7M).

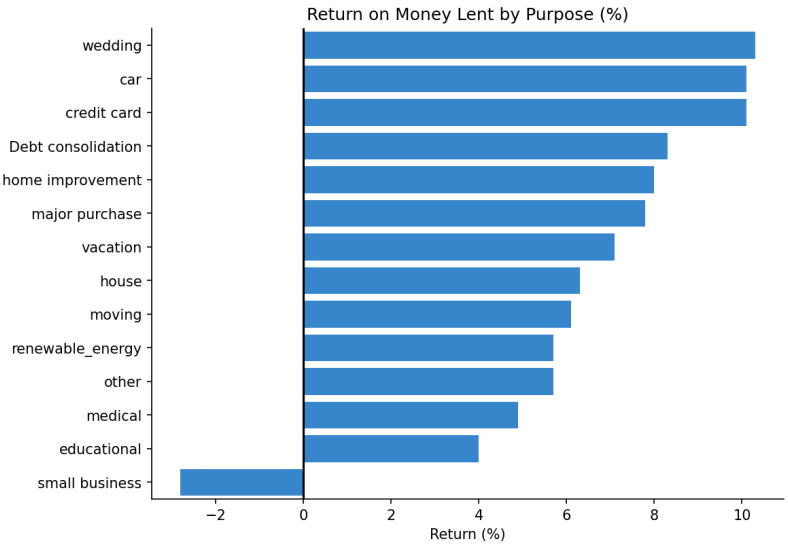



*Figure 23: Return on money lent by purpose*

**What it shows:** Small business is the only purpose that loses money (−2.8%): $22.8M lent and $22.1M returned. Wedding, car and credit card purposes return about 10%. 60-month loans return less than 36-month loans (7.2% vs 7.9%) despite far higher risk.

## 4.5 Geography

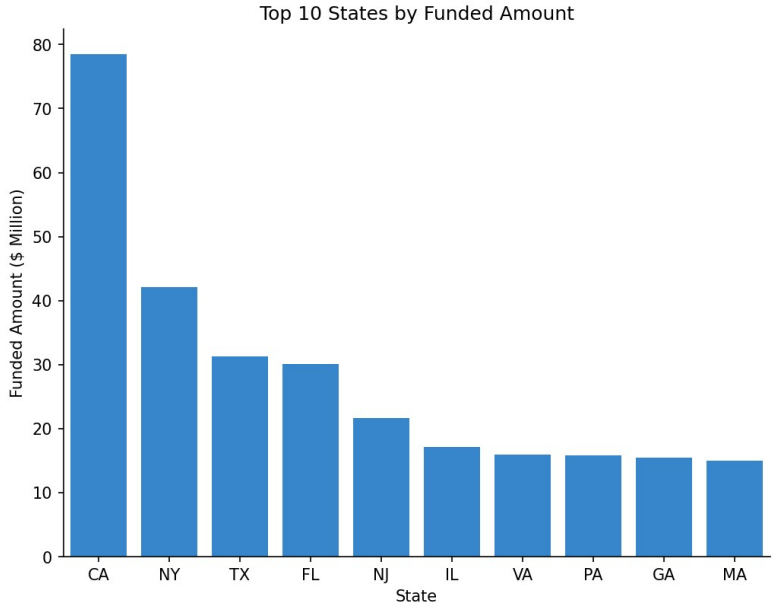



*Figure 24: Top 10 states by funded amount*

**What it shows:** California ($78M, 18% of all lending), New York, Texas and Florida together hold about 42% of the money lent.

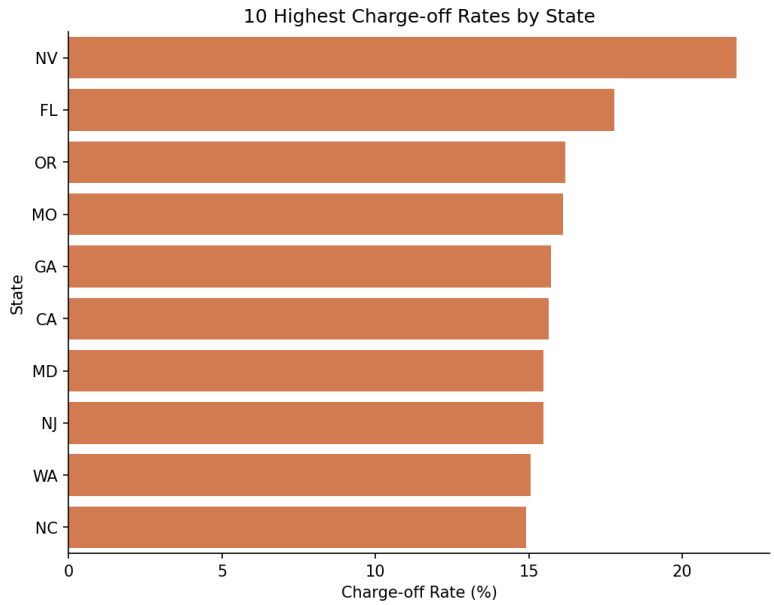



*Figure 25: Ten highest charge-off rates by state (states with 300+ finished loans)*

**What it shows:** Nevada (21.8%) and Florida (17.8%) are the worst. Florida matters most because it is large (2,691 finished loans) with a return of only 4.2%. California, the biggest market, is also above average at 15.7%. Texas (11.6%) is the safest large state.

## 5. Recommendations

| # | Recommendation | Evidence | Priority |
|---|---|---|---|
| 1 | **Tighten 60-month lending:** require a better grade or lower maximum amount, and re-price these loans. | 25.0% vs 10.7% charge-off; worse in every grade; lower return (7.2% vs 7.9%). | High |
| 2 | **Fix or restrict small business loans:** raise rates, cap amounts or ask for more documents. | 26.7% charge-off and −2.8% return – the only losing purpose. | High |
| 3 | **Focus growth on grades B and C** (and selected D); keep A for low-risk volume; limit F and G. | B and C give about half of net profit; A returns 4.5%; F/G have 32–33% charge-off. | High |
| 4 | **Add stricter checks on large loans** (above $15K): income and DTI limits. | Charge-off 17.1% ($15–20K) and 20.0% ($20K+) vs about 13% for smaller loans. | Medium |
| 5 | **Watch the quality of growth:** track each month's loans over time and set alerts for average loan size, rate and DTI. | Average loan +17%, rate 11.5% → 12.4%, December DTI +2.7% MoM. | Medium |
| 6 | **Review the verification process:** use risk-based checks rather than blanket verification. | Verified loans default more (15.6% vs 12.4%). | Medium |
| 7 | **Limit state concentration:** set exposure limits for California and review underwriting in Nevada and Florida. | Top 4 states = 42% of lending; NV 21.8% and FL 17.8% charge-off. | Medium |

| # | Recommendation | Evidence | Priority |
|---|---|---|---|
| 8 | **Extend the dashboard with a Risk page:** charge-off rate by grade, term and purpose, plus profit by grade. | These are the metrics that drive decisions but are not on the current pages. | Low |

# 6. Limitations and Next Steps

## Limitations

- The data covers **one year (2021)**, so long-term trends and seasonality cannot be confirmed. Loans issued late in the year have had less time to default.
- **Current loans are excluded** from risk and profit analysis. All of them are 60-month loans, so the 60-month charge-off rate is measured only on loans that have finished.
- Return is a **simple return** and recovery costs are not in the data, so true profit may be lower.
- The patterns shown are **associations, not proof of cause** (for example, the verification result).

## Next steps

- Add a **Risk & Profitability** page to the dashboard (see recommendation 8).
- Build a simple **default-prediction model** (for example logistic regression) using grade, term, loan amount, income and DTI.
- Track loan performance by **month of issue** once more months of data are available.